

Iptables: The Flexible Firewall Utility for Linux

This article covers the basics of Iptables and will add to your understanding of this firewall utility. After going through it, you can try your hand at mixing and matching different sets of rules as per your needs.



Iptables is a firewall utility built for Linux (2.4.x and later) operating systems. Whether you're a Linux techie or a systems administrator, it's very likely that Iptables can be of extraordinary use to you in some way or the other.

Iptables is used to set up, maintain and inspect the tables of IPv4 packet filter rules in the Linux kernel. You can define several different tables in the Linux system. Each table contains a number of built-in chains and may also contain user-defined chains. When a connection tries to establish itself on your system, Iptables looks for a rule in its list to match it to. If it doesn't find one, it resorts to the default action.

How Iptables works

When Iptables is able to match a connection that tries to establish itself on your system with a rule in its list, the action that takes place is called a target. A target can be a final policy decision for the packet, such as *Accept* or *Drop*. It can also be *Move the packet to a different chain for processing*.

A firewall rule specifies criteria for a packet and a target. If the packet does not match, the next rule in the chain is examined; and if it does match, then the next rule is specified by the value of the target, which can be the name of a user-defined chain or one of the special values like *Accept*, *Drop*, *Queue* or *Return*.

Iptables tables and chains

The Linux kernel uses the Netfilter facility to filter packets, allowing some of them to be received by or to pass through the system while stopping others. This facility is built into the Linux kernel, and has five built-in tables or rules lists, as follows.

- **Filter:** This is the default table for handling network packets.
 - **NAT:** This is used to alter packets that create a new connection and also used for Network Address Translation (NAT).
 - **Mangle:** This is for specific types of packet alterations.
 - **Raw:** This is mainly for configuring exemptions during connection tracking in combination with the NOTRACK target.
 - **Security:** This is used for Mandatory Access Control (MAC) networking rules, such as those enabled by the SECMARK and CONNSECMARK targets.
- Each table has a group of built-in chains, which correspond to the actions performed on the packet by Netfilter.

Filter table

This is the default table for Iptables. If you don't define any table, you use the filter table.